

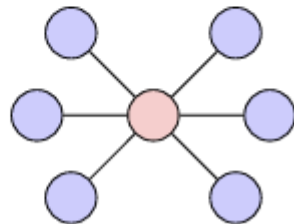
Social Network Analysis

Ch2: Network Models I

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The notion of centrality

- ▶ Identify the **most important nodes** in a graph given its **topology**
 - ⇒ **Not** based on the **nature** of the particular node
- ▶ Different definitions of **importance** give rise to different **centrality measures**
 - ⇒ Degree, closeness, eigenvector, betweenness, Katz
 - ⇒ They induce a **centrality ranking** on the nodes
- ▶ Centrality measures are **widely used**
 - ⇒ Targeted marketing
 - ⇒ Network vulnerability to attacks
 - ⇒ Epidemiology control
 - ⇒ Power in exchange networks



Degree centrality

- ▶ **Local** measure of the importance of a node within a graph
- ▶ Sum of the **weights of incident edges**

$$C_D(i) := \sum_{j|(i,j) \in E} A_{ij}.$$

- ▶ **High degree centrality** value of a given node
 - ⇒ The node has a **large number of neighbors**
 - ⇒ **Closely related** to its neighbors (in weighted similarity graphs)
- ▶ For directed networks, both in-degree and out-degree centralities
- ▶ Does not capture **cascade effects**
 - ⇒ I am more important if my neighbors are important

Closeness centrality

- ▶ How fast **information can spread** from one node to every other node
- ▶ Inverse of the **sum of the shortest path lengths s_G** to other nodes

$$C_C(i) := \left(\sum_{j \in V} s_G(i, j) \right)^{-1}.$$

- ▶ Nodes which are close to others on average have high centrality
 - ⇒ Such nodes may have more direct influence on others
- ▶ **Example:** Network of movie actors: two actors are connected if they work together
 - ⇒ Highest centrality 0.4143 for Christopher Lee; Entered into the Guinness Book of World Records in 2007 for most screen credits
 - ⇒ Lowest centrality 0.1154 for Leia Zanganeh, an Iranian film and theater actress
- ▶ Two limitations associated with closeness centrality
 - ⇒ Spans a very small dynamic range
 - ⇒ In disconnected graphs, centrality is zero for all nodes

Betweenness centrality

- ▶ Control on the **optimal flow** within a graph
 - ⇒ Nodes that fall within the **shortest path** of many pairs of nodes
- ▶ The **number of shortest paths** from j to k is σ_{jk}
- ▶ The **number of shortest paths** from j to k **going through** i is $\sigma_{jk}(i)$

$$C_B(i) := \sum_{\substack{j,k \in V \\ j \neq i \neq k}} \frac{\sigma_{jk}(i)}{\sigma_{jk}}. \quad (1)$$

- ▶ Look at the **shortest paths** for every two nodes distinct from i
 - ⇒ Sum the **proportion that contain node i**
- ▶ Betweenness is not a measure of connectivity.
 - ⇒ For the network of movie actors, highest centrality node is Spanish actor Fernando Rey, with centrality of 7.47×10^8
 - ⇒ He starred in many French, American and Spanish movies.

Interpretation in a social network

- ▶ What would **degree**, **closeness**, and **betweenness** centrality reveal?
- ▶ **Degree** $\Rightarrow$ Most friends $\Rightarrow$ Most **popular** person
- ▶ **Closeness** $\Rightarrow$ Can quickly reach the whole group (directly or indirectly)
 $\Rightarrow$ Relevant if we want to **quickly spread information** in the network
- ▶ **Betweenness** $\Rightarrow$ Power in the transmission of information
 $\Rightarrow$ Relevant if we want to **influence communication** between groups
- ▶ All of them are right, they just reveal different features

Eigenvector centrality

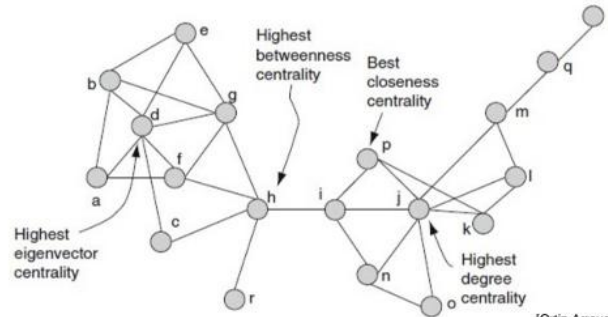
- ▶ **Linear combination** of the centrality of the neighbors
⇒ **Few important** neighbors can weigh more than **many unimportant** ones

$$C_E(i) := \frac{1}{\lambda} \sum_{(i,j) \in E} A_{ij} C_E(j),$$

- ▶ This definition leads to an **eigenvector** problem
- ▶ C_E is the **dominant eigenvector** of the adjacency matrix
⇒ Perron-Frobenius guarantees **all positive elements**

Comparing centrality measures

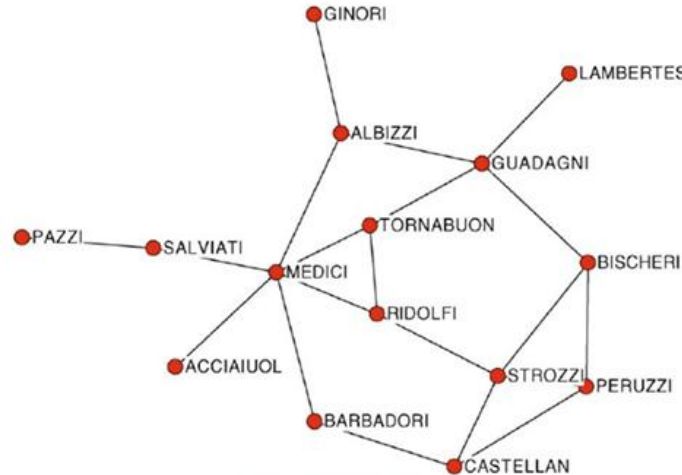
- ▶ Different measures target different notions of importance
- ▶ In a friendship network, degree centrality would correspond to who is the most popular kid.
- ▶ Closeness centrality would correspond to who is closest to the rest of the group,. Useful for spreading information
- ▶ Betweenness would tell us about graph “cut points”, edges whose deletion will cause multiple connected components



[Ortiz-Arroyo 10]
Ortiz-Arroyo, Daniel. "Discovering Sets of Key Players in Social Networks." Chapter 2 in *Computational Social Network Analysis: Trends, Tools and Research Advances*. Springer, 2009. © Springer. All rights reserved. This content is excluded from our Creative Commons license. For more information, see <https://ocw.mit.edu/help/faq-fair-use/>.

Centrality and power in networks

- ▶ Why were the Medicis the most influential family in 15th c. Florence?
- ▶ Political and friendship structure [Padgett & Ansell 93]



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Source: Stephen P. Borgatti. "Centrality and Network Flow," *Social Networks* 27 (2005): 55–71.

- ▶ Highest betweenness centrality by far
⇒ Part of many deals between families supported by marriage linkages

Katz centrality

- ▶ Degree centrality only depends on the one-hop neighbors of a node
 - ⇒ One-hop neighbors more relevant than two-hop neighbors
 - ⇒ But indirect relationships are still relevant
- ▶ $[\mathbf{A}^k]_{ij}$ contains the number of walks from i to j of length k
 - ⇒ Consider the degrees of $\mathbf{A}$, $\mathbf{A}^2$, ..., with a discount factor

$$C_K(i) := \sum_{k=1}^{\infty} \alpha^k \sum_{(i,j) \in E} [\mathbf{A}^k]_{ij} = [(\mathbf{I} - \alpha \mathbf{A})^{-1} \mathbf{1}]_i$$

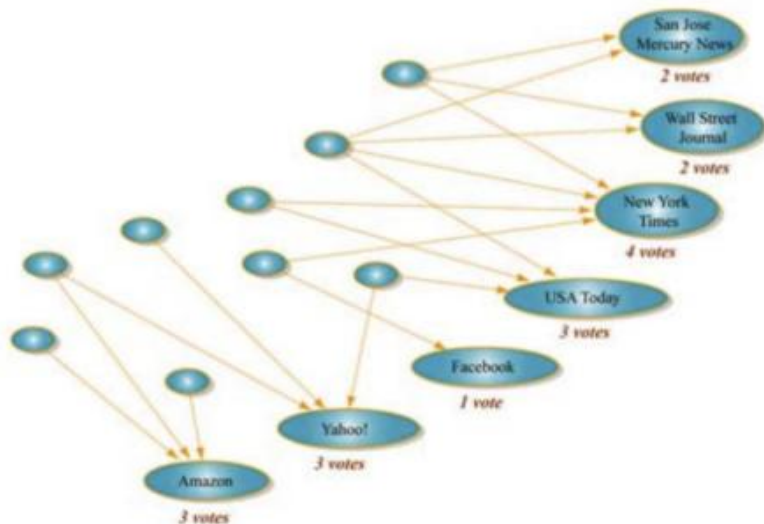
- ⇒ where α is small enough to ensure that the series converges
- ▶ Katz centrality as a hybrid between degree and eigenvector
 - ⇒ Parameter α controls this transition

Centrality and web search

- ▶ When you go to Google and type MIT
 - ⇒ First result is the home page of MIT
 - ⇒ How does Google know that this was the best answer?
- ▶ Problem of **information retrieval**
 - ⇒ Search data repositories in response to keyword queries
 - ⇒ Classical approach has been **textual analysis** ⇒ No link structure
- ▶ Links that point to the webpage ⇒ **authority** of a page on the topic
 - ⇒ First, collect a large sample of pages relevant to a topic
 - ⇒ Then, look at the number of in-links (score) from these pages

Link structure

- For a query like "newspaper", the most important page is less obvious



Leskovec, Jure, Anand Rajaraman, and Jeffrey David Ullman. *Mining of Massive Datasets*. Cambridge University Press, 2019.
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- Mix of newspapers and other pages that always receive links

Page Rank

- ▶ Multi-billion query-independent idea of Google
- ▶ Each node (or page) is **important** if it is **cited by other important pages**
- ▶ Each node j has a centrality value (**PageRank value**) $w(j)$
 - ⇒ Function of the centralities of his (incoming) neighbors
 - ⇒ Similar to eigenvector centrality

$$w(j) = \sum_i \frac{w(i)}{d_{out}(i)} A_{ij}$$

- ▶ Dividing by degree dilutes the importance of pages linking to many nodes
- ▶ In matrix notation $\mathbf{w}^T = \mathbf{w}^T \mathbf{P}$ where $P_{ij} = A_{ij}/d_{out}(i)$
 - ⇒ Note that $\sum_j P_{ij} = 1$

Page Rank

- ▶ This defines a **random walk** on the nodes of the network
 - ⇒ Walker starts from a node chosen uniformly at random
 - ⇒ Walker moves out choosing uniformly among the out-links
- ▶ **PageRank** is the **limiting probability** of the random walk
 - ⇒ But, **dangling ends** may cause the walk to get trapped
- ▶ We allow random walk to **teleport** with probability $1 - s$

$$\mathbf{w}_{k+1}^T = s\mathbf{w}_{k+1}^T \mathbf{P} + \frac{1-s}{n} \mathbf{1}$$

- ▶ This is a simpler version of **PageRank** ⇒ More tricks in practice

Eigenvalues and eigenvectors of graph matrices

- ▶ Vector v is an **eigenvector** with **eigenvalue** λ if

$$Mv = \lambda v.$$

For any **symmetric real $n \times n$ matrix M** :

- ▶ If v and w are eigenvectors with distinct eigenvalues, then v and w are orthogonal.
- ▶ If v and w are eigenvectors corresponding to the same eigenvalue, then for any scalars a and b , $av + bw$ is an eigenvector with the same eigenvalue as v and w .
- ▶ M has a full orthonormal basis of eigenvectors $v_1, v_2, \dots, v_n$. All eigenvalues and eigenvectors are real.
- ▶ M is diagonalizable. That is,

$$M = V\Lambda V^T,$$

- ▶ V : matrix with n orthonormal eigenvectors as columns
- ▶ Λ : diagonal matrix with eigenvalues on diagonal
- ▶ $M = \sum_i \lambda_i v_i v_i^T$ (note that $VV^T = I$).

Eigenvalues and eigenvectors of graph matrices

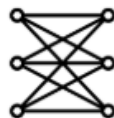
- Complete graph K_5 :

$$\lambda(A) = \{4, -1, -1, -1, -1\}$$



- Bipartite graph $K_{3,3}$:

$$\lambda(A) = \{3, 0, 0, 0, 0, -3\}$$



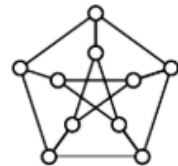
- Ring graph P_5 :

$$\lambda(A) = \left\{2, \frac{-1+\sqrt{5}}{2}, \frac{-1+\sqrt{5}}{2}, \frac{-1-\sqrt{5}}{2}, \frac{-1-\sqrt{5}}{2}\right\}$$



- Peterson graph:

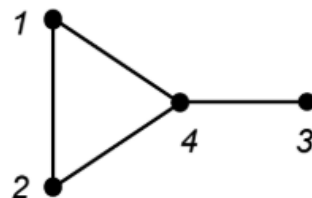
$$\lambda(A) = \{3, 1, 1, 1, 1, 1, -2, -2, -2, -2\}$$



Graph Laplacian

- ▶ Vertex degrees often stored in the diagonal matrix $\mathbf{D}$, where $D_{ii} = d_i$

$$\mathbf{D} = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}$$



- ▶ The $|V| \times |V|$ symmetric matrix $\mathbf{L} := \mathbf{D} - \mathbf{A}$ is called **graph Laplacian**

$$L_{ij} = \begin{cases} d_i, & \text{if } i = j \\ -1, & \text{if } (i, j) \in E \\ 0, & \text{otherwise} \end{cases}, \quad \mathbf{L} = \begin{pmatrix} 2 & -1 & 0 & -1 \\ -1 & 2 & 0 & -1 \\ 0 & 0 & 1 & -1 \\ -1 & -1 & -1 & 3 \end{pmatrix}$$

- ▶ Variants of the Laplacian exist, with slightly different interpretations
 - $\Rightarrow$ **Normalized Laplacian** $\mathbf{L}_n = \mathbf{D}^{-1/2} \mathbf{L} \mathbf{D}^{-1/2}$
 - $\Rightarrow$ **Random-walk Laplacian** $\mathbf{L}_{rw} = \mathbf{D}^{-1} \mathbf{L}$

Code

- <https://colab.research.google.com/drive/1nx8ZfVg0KCQU15pCBavmH4g-wbfUkt5Z?usp=sharing>